

Xingrui Chen

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EDUCATION

Duke University

Aug 2025 - May 2027 (Expected)

M.S. in Mechanical Engineering, Robotics Concentration

GPA: 3.85/4.00

Coursework: Robotic Manipulation, Computer Vision, Data Science & Machine Learning.

Hefei University of Technology

Sep 2021 - Jun 2025

B.Eng. in Mechanical Engineering

GPA: 3.44/4.00

Honors: First Prize, China Formula Student Competition; First Prize, 3D Modeling Competition.

TECHNICAL SKILLS

Robotics Software: Python, MATLAB, ROS 2, MoveIt, TF2, URDF, RViz, Gazebo, Isaac Sim.

Planning & Control: Inverse kinematics, trajectory generation, workspace calibration, PID tuning, visual feedback.

Perception & Design: RGB-D point clouds, DBSCAN, image registration, SolidWorks, CATIA, ANSYS, 3D printing.

WORK EXPERIENCE

Duke University | DexLab

Jan 2026 - Present

RESEARCH & PROJECT EXPERIENCE

SO-101 Robotic Painting | Motion Planning & Visual Feedback

Apr 2026 - Present

- Built an image-to-motion pipeline with boundary tracing, contour-loop filling, and quadratic Bézier strokes; organized color/tool batches and generated joint commands for physical execution on a LeRobot SO-101.
- Calibrated a **120 × 120 mm** workspace with a **49-point** grid; mapped paper coordinates to joint configurations through bilinear interpolation for tool lowering, stroke execution, and inter-stroke transfers.
- Implemented visual checkpoints after up to **3 strokes**, using image registration and stroke differencing to update future XY commands with bounded corrections constrained by workspace limits.
- Tuned five joints and reduced aggregate joint-angle **MAE by 47%** (0.527° to 0.279°) and **P90 by 67%** (1.618° to 0.527°), pooling 120 positioning measurements across the five joints.

Dual-Arm Autonomous Manipulation | ROS 2 & MoveIt

Sep 2025 - Dec 2025

- Co-developed a perception-planning-execution pipeline for two Kinova Gen3 Lite arms, integrating RGB-D point-cloud filtering, TF2 transforms, DBSCAN block detection, and YAML board-state/action interfaces.
- Built MATLAB kinematic models and inverse-kinematics reachability checks for all **9 board cells**; confirmed both arms could reach every cell and characterized the fully overlapping task workspace.
- Integrated workspace partitioning and shared-zone mutual exclusion into coordinated pick-and-place execution; validated full-game operation in Gazebo and on hardware, with no arm-arm contact observed in the reported trials.

CEER | Humanoid Simulation & Hardware Validation

Jan 2026 - Apr 2026

- Built room-scale Isaac Sim test scenes with randomized furniture positions (± 0.5 m), heights (0.5-0.7 m), and three boxes; tested humanoid grasping, transport, and placement for system evaluation.
- Contributed to end-effector design and 3D printing; completed **10+ hours** of hardware testing and data collection for the humanoid loco-manipulation system.

Robot-Assisted Laser Lithotripsy | Control & Experimental Validation

Jan 2026 - Present

- Integrated a DOBOT Nova 5 with a Ho:YAG laser testbed and programmed two- and four-step fiber-positioning protocols; compared them with fixed-standoff control at **0.4 J, 20 Hz** in underwater ablation experiments.
- Developed MATLAB segmentation of **512³-voxel OCT scans** to quantify crater geometry; four-step control increased crater volume by **44.5%** at 320 pulses and 0.75 mm initial standoff versus fixed control ($p = 0.00173$).

Humanoid Waist-Joint RV Reducer | Mechanical Design & FEA

Oct 2024 - May 2025

- Designed a two-stage RV reducer in SolidWorks for a **220 N · m** output target within a **180 mm** diameter limit; selected a coaxial frameless motor and evaluated gear contact stress and deformation in ANSYS.

Formula Student | Aerodynamics & Bodywork Team Lead

Sep 2022 - Oct 2024

- Led **5 teammates across 2 seasons** to design and fabricate carbon-fiber wings; improved simulated downforce-to-drag ratio by **16%** (2.5 to 2.9) through CATIA design iterations and STAR-CCM+ CFD.

PUBLICATIONS

- Ni, W., Li, J., **Chen, X.**, Li, H., Alessio-Bunnell, C. A., Dong, Y., and Cheng, X. "Any-ttach: Quick End-effector Swapping Enables Manipulation Dexterity with Simplicity." *RSS Dexterous Manipulation Workshop*, 2026.
- Luo, X., **Chen, X.**, Yin, X., Wu, H., Xia, B., Chen, Z., Li, J., Chen, B., and Cheng, X. "CEER: Compliant End-Effector and Root Control as a Unified Interface for Hierarchical Humanoid Loco-Manipulation." *IEEE/RSJ IROS*, 2026.
- Chen, X.**, Ma, X., Ni, W., and Cheng, X. "Learn, Imagine, and Paint: Aesthetic Cognition as Guidance for Robotic Painting." *RSS Workshop on Embodied Arts*, 2026.
- Chen, X.** "Analysis and Improvements of a Front Wing for Formula SAE Car." *International Conference on Computing Innovation and Applied Physics (CIAP)*, 2024.